Foundation for Virtual and Augmented Reality Systems

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Week – 03

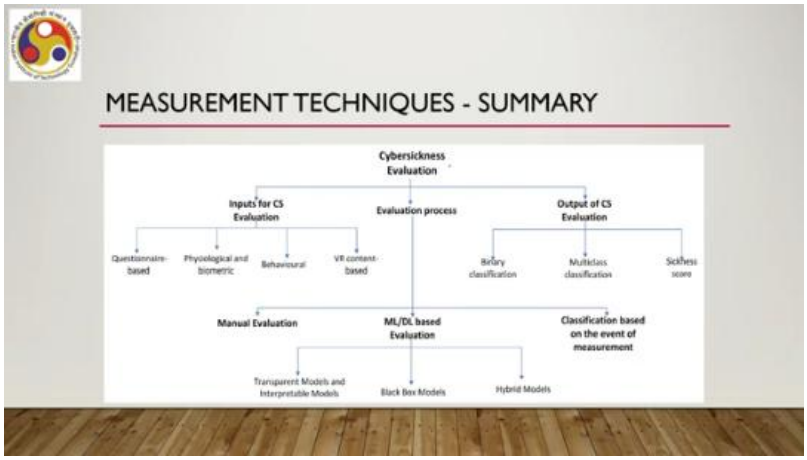
Lecture – 12

Lec 12: Cybersickness

Hello and welcome to the NPTEL MOOC's course on Foundations of Virtual and Augmented Reality Systems. We are currently discussing cybersickness, which is lecture number 6 or topic number 6. We will continue our discussion here. So, so far whatever we have discussed, if we just quickly recollect, we have discussed the causes of cybersickness, different factors, and then we started our discussion on measurement and evaluation of cybersickness, how do we measure it.

So, partly we have covered some of the evaluation techniques. In this lecture, we are going to continue our discussion on evaluation techniques. Also, we will conclude by having a discussion on the mitigation strategies that can be adopted both from the developers' point of view as well as users' point of view to reduce or eliminate cybersickness. So if you recollect, we started our discussion on evaluation strategies.

There are a large number of strategies. We can organize them in this form as shown here. So broadly, we can have three aspects of evaluation: that is the input that we provide, the process itself, and the output. We talked about different inputs that can be collected, namely questionnaire-based subjective inputs, physiological and biometric inputs, behavioral inputs, and content-based inputs. We also started our discussion on the evaluation process.



We already covered manual evaluation and started discussing MLDL-based evaluation where we talked about transparent models. Now, let us continue our discussion and talk about other models: interpretable black box and hybrid models. Also, let us talk about output, what kind of output we can expect from the evaluation process. So, interpretable models are similar to the transparent models that we discussed in the previous lecture.

They offer more flexibility in terms of customization, the way we use the model, and provide more precise details about the relationships between the input that we provide and the output that we get. Those are the properties of interpretable models. Several interpretable machine learning algorithms were used to measure CS. Some of those related are, some of those are CN2 rule induction. Bagged decision trees, Naive Bayes classifiers, KNN, LDA, LSTM regression analysis, and so on.

Sometimes combinations of different ML techniques were also used to identify and provide insight into cyber sickness. For example, explainable boosting machine or EBM, decision tree, and logistic regression were used in combination to get some insight where biophysiological and subjective measurements were used as input. Then come the black box models. These are models

that cannot be easily understood or explained, and their decision-making process is shrouded in mystery. So, essentially we do not know how they work; we provide input and we get some output, but it is very difficult to understand how the output is generated based on the input, how the input and output are related, unlike the transparent or interpretable models. So, essentially we assume the models to be a black box which just takes input and produces some output. That is why these types of approaches are called black box models.

Some of the models studied were LSTM, CNN and SVM models, support vector machines. In a study, these models were compared and LSTM was found to achieve the best results with respect to CS measurement. 3D CNN along with a multimodal diffusion method were explored, which integrated optical flow disparity and saliency features, some of the causes of cyber sickness. Then CNN and LSTM were used to estimate cognitive states based on brain signals.

and their relationship with CS levels, how brain signals affect CS. So, these are some of the works that used black box models. Generally, use of black box models leads to high prediction accuracy as compared to the other models that we have seen namely the transparent and interpretable models. However, they require huge amounts of data and accordingly good amounts of processing power.

So, we require computational resources and data to be able to use black box models, which is not that much required for other types of models. And also as we already have discussed these are difficult to interpret we do not know how they work or why they work. So, that is one of the shortcomings we can say of these models. But if we are only concerned about the output, then generally they tend to give better prediction accuracy as compared to other models. The third category of approach concerns hybrid models.

Hybrid models, as the name suggests, are combinations of different ML models designed to provide high accuracy and transparency in their decision-making process. So, when we are combining multiple ML models, we call them hybrid models. For example, a method that used a hybrid model used a deep embedded self-organizing map or DESOM and an autoencoder designed from EEGNET, a specialized CNN for EEG prediction. Now, that combination was used to predict cybersickness. Another study employed a voting classifier for binary and three-level classification of cybersickness.

which used four basic classifiers namely random forest, KNN, logistic regression and multilayer perceptron neural network. Again, combination of different basic models. Hybrid models can potentially combine strengths of both transparent and black box models by providing good predictive accuracy and some degree of interpretability. So, when we are combining models, we may combine transparent models with black box models.

Now, so partly we are able to understand why it works because of the presence of transparent models and we are also getting high accuracy because of the nature of black box models. So, it combines the best of both worlds. They can be useful in situations where accuracy as well as interpretability are essential such as medical diagnosis. So, sometimes

in some application domains, we require both high accuracy as well as transparency. In such cases, hybrid models offer a better solution as compared to only one type of model, either transparent models or black box models.

Then there is a third category of evaluation technique that is classification based on the event of measurement. What does it mean? Most work in the literature that we have covered so far, which we have discussed earlier, primarily focused on the detection of CS, that means after it happened. So, whatever work we have discussed earlier takes input after the experience took place. Based on the experience, the input is generated and that input is fed to the system to detect if cyber sickness has happened or not.

Now, there are very few studies on predicting CS, that is before it happens. That is also important. So, before it happens, if we know that cyber sickness is going to happen, then we can take some mitigation measures as we shall see in the next part of the lecture. But most of the works ignore that part. They generally focus on detection rather than prediction.

However, that is equally important. Some works employed physiological signals and video features to predict CS. Limited research exists on the ability to anticipate upcoming CS incidents by employing pre-VR immersion baseline measurements. That has also happened that was done, but on a very limited scale. Then there are few ML research studies that were focused on predicting susceptibility or impending CS episodes.

Few works are there, but more may be required. We have covered manual methods to evaluate, automated methods mostly related to detection, then classification based on event measurement which is primarily geared towards prediction. Let us now move our attention to the third part of the evaluation that is output. What kind of outputs are generated when the models complete their work? So, output of self-report questionnaires for CS are generally scores or ratings.

So, these questionnaires provide rating scales, users are supposed to provide those ratings. So, the outputs are obviously ratings. Now, based on the scores or ratings, we can assess the severity of CS. For ML-based evaluation, machine learning-based evaluation, the classification labels or classes can be one output. It can also be a sickness score.

Now, classification indicates likelihood or severity of CS. For example, low, moderate, acute or sickness, no sickness, this type of classification we can think of. Whereas, sickness score quantifies severity of CS experience. Generally, the sickness score predictor uses a

regression model. So, we can have either way the output that is either a class or a numerical value called the sickness score.

So, with that we have come to the end of our discussion on evaluation or measurement techniques. Now, let us move our focus to the third aspect of cyber sickness, that is mitigation. So far we have seen the causes, we have seen how to measure it, let us see how we can reduce or eliminate cyber sickness. If it is not possible to completely prevent CS, which is generally the case, it is not possible to completely prevent, we can at least try to lessen its impact by taking some measures. Now mitigation strategies often come at the cost of presence, that is very important, that if we follow some mitigation strategy then it may affect presence or user's ability to perceive

in the perceive the virtual world and that is where the challenge lies. We wish to develop strategies without compromising user experience. So, if presence is affected certainly it has an impact on user experience and we wish to avoid that to the extent possible. So, there is a trade-off which one should be compromised to what extent, whether we completely avoid cyber sickness by reducing presence to the maximum level or we can balance the trade-off.

Now, when we are talking about mitigation strategies, we can look at it from two points of view. One is the developer's perspective, that is what the developers can do, want to do, or the way the developer sees the problem, how developer can address it. The other thing is the user's perspective. What users can do, no matter how good a solution is, if the user is... not doing it properly or using it as expected, cyber sickness may still happen.

So, developers might have all the good intentions, but usage may ruin the potential. So, both perspectives are important, developers as well as users. Let us have a look at both perspectives. From the developer's point of view, there are many strategies that can be adopted while building the system.

One of these is the FOV or field of view adjustment. Now, FOV is, adjusting FOV means the amount of virtual environment visible at any moment; we can adjust that. So, essentially by adjusting the amount of virtual environment visible at any moment, we can mitigate the onset or severity of cyber sickness. Restricting FOV reduces peripheral optical flow. Now, if you can recollect from the earlier part of the lecture, peripheral optical flow has an impact on our visual motion perception and which in turn leads to

the occurrence of cyber sickness. So, by restricting FOV, if we can reduce this peripheral optical flow, then we can, ideally, reduce cyber sickness. It has been found that adjustment

of FOV is effective across genders without any negative impacts. But the drawback is it can reduce presence or sense of presence and impair performance in visual tasks because certain things are restricted.

So, it may lead to the reduction in user experience, degradation in user experience. Now, there are various strategies to adjust FOV. Let us have a quick look and a brief understanding of each of those strategies. First thing is symmetric mask.

So, we can occlude the periphery in reaction to translational and/or angular velocity, a technique called symmetric mask. So, this is the classical approach, but it can affect presence. Then we have asymmetric mask. We obscure periphery with higher optical flow during virtual locomotion while leaving regions with lower optical flow visible. So, when there is higher optical flow in the periphery, we obscure that periphery, whereas regions where there is lower optical flow, we make it,

or keep it visible. So, this can also be considered to be an adaptive restrictor. So, it adapts based on optical flow. Then we have asymmetric mask which is a side restrictor. What happens here is we obscure only one side of the periphery during virtual rotation and laterally shift the center of restriction in the direction of the turn. So, it is a controlled way of restriction. This is a side restrictor. We can also have a ground visible restrictor. So, the ground plane remains visible during movement.

So, these are the broadly FOV adjustment techniques that A developer can adopt to improve or to reduce the chances of cyber sickness while the user is experiencing it. However, some of them may lead to issues with presence. Another technique can be dynamic visual blur. So there can be two types of such blur effects.

Depth of field blur makes use of well-known depth cues in human vision and peripheral blur. So dynamic non-salient region blurring. Blurring means obscuring or reducing the clarity. So, we can have two types of such blur effects: depth of field blur and peripheral blur. This can be another way of reducing the possibility of cyber sickness.

Rest frames or the use of rest frames is another technique. Rest frames are fixed elements within the virtual environment that remain stationary relative to the real world regardless of the user's movements. So, rest frames attached to the head reference frame are known as independent visual backgrounds. For example, clouds, grids, and a virtual nose. So, these are some of the examples of rest frames which remain stationary relative to the user's movement.

Optical flow manipulation is another technique. It refers to visually mixing optical flow patterns that are in the reverse direction of virtual visual motion. We can also reduce optical flow in peripheral motion as we have seen earlier. A restricted FOV aligned with optical flow can be another technique. Reduce perceived optical flow by means of geometrical simplification of the virtual environment and other techniques.

So, some of the techniques we have already discussed as part of earlier techniques, some more related to optical flow manipulation can be used, and all these techniques essentially try to manipulate or change the way optical flow happens, which in turn may lead to a reduction in or reduction in the possibility of cyber sickness. We can also have hardware level alterations done to reduce the effects of cyber sickness. We can modify the hardware to propose totally new hardware; both are possible.

So, we can modify it to different degrees, or we can propose completely new hardware to reduce or eliminate cyber sickness. One such example can be sparse peripheral display. Here we are expanding HMD periphery with colorful LED matrix. So, that is a hardware addition.

Then we can have miniature LED screens placed in the HMD. Noble camera setup twin cam reduces motion blur and CS. Hybrid 360 panoramic video camera rig can be used to reduce CS. Phantom legs was another technique developed. It applies alternating haptic cues synchronized to users' footsteps using additional hardware.

It was found that this technique managed to reduce CS to a great extent. Floor vibrations, another technique developed and tried out. It tries to match with vibrations of virtual environment. utilizes vibration driven whole body tactile feedback. So, again, these are hardware-level alterations that are done to prevent or reduce the effect of cyber sickness.

Sometimes through hardware we can provide sound and vibration simultaneously. So, simultaneous presentation of sound and vibration was also tried. For example, the walking vibe system where two small vibration motors behind the ears for vibrotactile feedback were used for walking experience. We can have modified motion platforms.

A study had shown that short bursts of rotational acceleration to match the visual forward acceleration was quite effective in reducing CS. So, accordingly we can modify the motion platforms to have this effect. Those were things that we can do at the hardware level. Also, we can have locomotion techniques modified or manipulated or

new techniques proposed to reduce the effect of cyber sickness. So, locomotion techniques can play a crucial role as we have seen earlier in the discussion on causes. They can play a crucial role in preventing cyber sickness. Say, for example, we can come up with novel walking methods rather than the natural walking method or we can have teleportation over steering-based locomotion. We can have dynamic steering speed to skip certain locations when going over long distances.

So, to reduce the exposure or duration of virtual exposure. Then viewpoint snapping is another technique we can use that is rendering moving pictures in discrete chunks during rotation. And a modification to the locomotion technique. We can also provide invisible ramps on top of stairs when traveling through stairs. So, these are some of the interesting ways, novel approaches to manipulate locomotion while letting the user

move from one point to another in a virtual world, which potentially can lead to a reduction in cybersickness. So, those are related to the developer's perspective. Equally important is the user's perspective, what the users can do or ideally how the user should behave to reduce the effect. As I said before, only developing a good system need not lead to the reduction in CS because even if there are features or there are modifications, alterations, novelty embedded in the system by the developer.

Users can ignore those; users can try to use it the way he or she likes, thus nullifying the potential benefits. So, users also have to take a little bit of caution while using such systems. So, what can users do? Now, the subjectivity involved in CS makes it a very complicated problem to deal with. Now, different individuals can have varied degrees of susceptibility towards CS.

So, a one-size-fits-all approach may not work. Developers can come up with some guidelines, but it should be noted that these guidelines or these modifications, alterations at the system level or the content level may not be applicable to everybody because it is very much subjective. Interventions only from the developers' end might not be enough because it is subjective. So, users also need to take some countermeasures on their own.

So, how can it be done? We already know it is well known, well established that increased exposure time in the virtual environment leads to severe symptoms of CS. So, time is very important as we have seen earlier while discussing the causes. Therefore, it may be suggested that to avoid continuous exposure, for longer periods of time in the virtual environment.

So, users can be suggested to avoid continuous exposure or long-duration exposure to the virtual environment. Studies have suggested that the maximum duration of a VR session should be between 55 to 70 minutes. Note that this number may vary depending on the kind of device you are using, the kind of content that you are experiencing, and your other factors. But in general, for the majority of users who are likely to suffer from cyber sickness, between 55 to 70 minutes of exposure time may be considered safe. Beyond that, it is risky.

So, users may take things accordingly. What can they do? Habituation. So, the intensity of cyber sickness gradually decreases after repeated experience in the virtual environment; that is called habituation.

So, repeated short-duration exposure until the undesirable symptoms appear may be tried. Try to get habituated with the system through repeated short-duration exposures. This can be followed by repeated longer-duration exposures after full recovery from cyber sickness symptoms during the earlier phases of habituation. So, a two-stage approach can be adopted here. One is that users can first try with short-duration repeated exposures.

This can be followed by longer-duration exposures. There can be other measures that users can take on their end. Say, for example, slow diaphragmatic breathing. So, by controlling their breathing patterns, they can, to a large extent, reduce the possibility of cyber sickness. Controlled breathing to delay the onset and/or reduce the severity of cyber sickness can be tried by the users.

Other possible strategies that users can adopt: playing pleasant music in the background can be one strategy. Ensuring adequate external air flow can be another strategy. So, proper air flow, playing background music, these can calm the mind, these can soothe the mind and these can lead to reduction in cybersickness. So, as we have seen, many things can be done, some from the developer side, system side can be done.

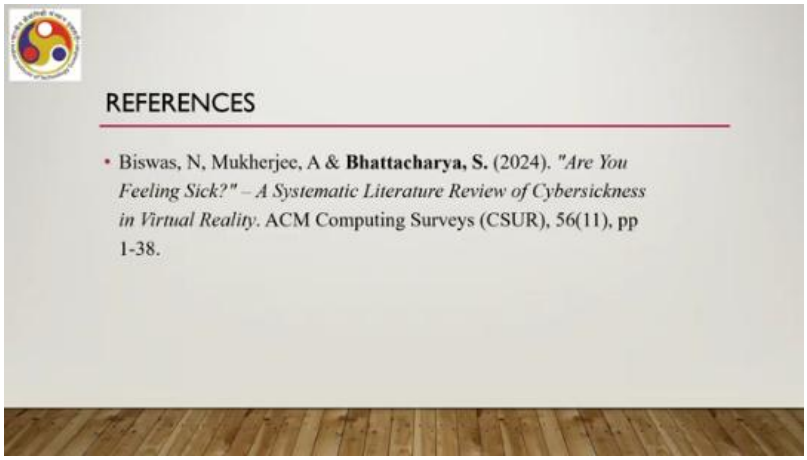
Also, it is advisable that users also take some measures proactively so that the effects of cybersickness can be reduced. With that, we have come to the end of our discussion on cybersickness. What we have covered so far, let us quickly summarize. So, we talked about cybersickness. This is an important issue in virtual reality particularly and we should try to reduce it or eliminate it altogether as developers.

We talked about various causes, broadly two types, internal and external. Internal are subjective and relate to age, gender and so on. External can be hardware related, can be

content related. We have also talked about how to measure cyber sickness, evaluation strategies, where we talked about the kind of input to the measurement process, the process itself, which can be manual or automated through machine learning approaches and the output that can be classes or CS scores or numerical values.

Then we talked about some mitigation strategies both from the developer's point of view and user's point of view. From the developer's point of view at a system level, many things can be done including alterations in hardware, However, as cybersickness is very much subjective, a one-size-fits-all approach or mitigation strategies may not work for everybody or even the majority. So, we have to be very careful while using the system. As users, we should also take some measures to reduce the effects.

So with that, we have come to the end. Next, we are going to talk about the hardware and software components and overview of the virtual reality and augmented reality technologies. The topics or the content that we have discussed in this lecture on cybersickness were mostly taken from this reference. If you are interested, you may go through it.



You will get to know all these topics in more detail. So, I hope you enjoyed the content and look forward to meeting you all in the next lecture on hardware and software aspects of the technology. Till then, goodbye and thank you. Happy learning.